Syracuse University
School of Information Studies

Week 7
Visualization with Maps

Dr. Preeti Jagadev

To make a **data-based** map (like a map that shows population, sales, rainfall, etc.), you always need three types of data:

- **Location Data:** This tells the map the place you are talking about. **Examples:** Latitude/Longitude, street address, ZIP code, city name, state name.

- **Attribute Data:** This is the actual thing you want to map. **Examples:** Population, number of sales, average temperature, land use.

- **Identifier / Key Field:** This is like a label that makes sure the right data goes in the right place.

Geographic Information System (GIS)

- It is a tool that makes and analyzes maps.
- It capture spatial (location-based) data, stores geographic data, and visualizes data on maps.
- A Geographic Information System helps answer questions like:
  - Where are certain events happening?
  - How are locations connected?
  - What patterns exist across regions?

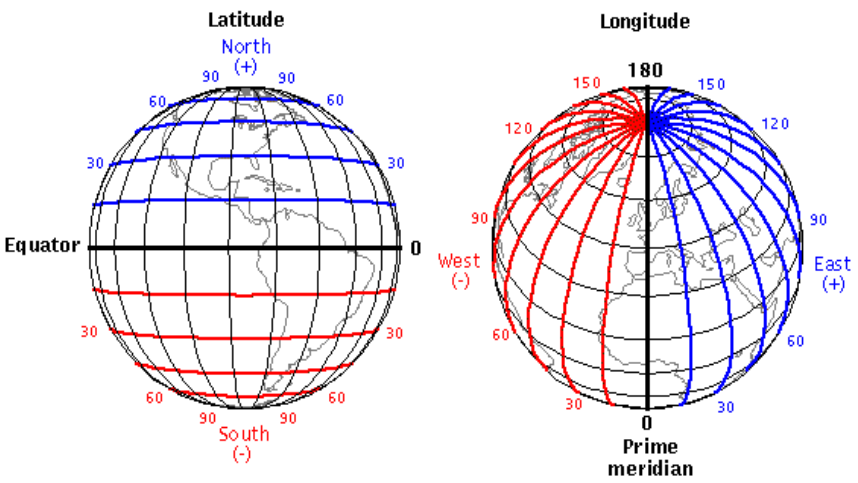
Geographic Coordinate Systems

Latitude (side-to-side lines):

- Latitude lines go horizontally around the Earth, like belts.
- They are parallel circles (never meet).
- 0° latitude is the Equator (the middle of the Earth), +90° latitude is at the North Pole, –90° latitude is at the South Pole.
- Each latitude line makes a full circle around the Earth.

Longitude (up-and-down lines):

- Longitude lines go vertically from the North Pole to the South Pole. They are not parallel; they meet at the poles.
- Each longitude line is a half-circle from pole to pole.
- The System Together: You need both a Latitude and a Longitude number to find any specific point on Earth.



- When plotting spatial (geographic) data on a 2D map, the x-axis defines the **longitude** of a location, and the y-axis represents its **latitude**

Map Projections

- **Map Projections:** How we take the round, three-dimensional Earth and draw it on a flat, two-dimensional map.

The Problem:

- **Distortion:**
- The main problem is that you cannot perfectly flatten a sphere.
- All map projections create distortions.
- Every single one has some type of error, whether in the shape, size, distance, or direction of landmasses.

Four Major Classes of Spatial Data

- The fundamental ways we represent real-world locations and objects in a digital map or Geographical Information System (GIS).

1. Point:

- A single, individual location, like a dot on a map.
- **Examples:** The location of a single tree, a street light, or a specific store.

2. **Line:**

- An ordered set of connected points. It has length but no area.
- **Examples:** A road, a river, or a hiking trail.

3. **Polygon:**

- An area that is completely enclosed by a set of lines. It has both length and area.
- **Examples:** A lake, a city boundary, country boundary, or the outline of a building.

4. **Raster or Grid:**

- A collection of locations organized into a regular, rectangular grid (like a checkerboard).
- **Examples:** A satellite image, a photo, or a map showing elevation or temperature where each grid cell has an elevation or temperature value.

Geo Encoding

The process of translating a street address that a person can read into precise geographic coordinates (latitude and longitude) that a computer can read and map.

Here are the details:

- **The Input (What you start with):**
  - A regular human-readable address, like a mailing street address (e.g., "1600 Pennsylvania Avenue, Washington, D.C.").
- **The Process (The conversion):**
  - Geo Encoding takes that address and searches a database to find its exact location on the Earth.
- **The Output (What you end up with):**
  - **Geographic coordinates** (a pair of numbers: **latitude** and **longitude**). These numbers tell a map where to place a pin.

How it's done:

- This conversion is usually performed using a specialized computer program, often called an **API** (Application Programming Interface).
- **Examples of these APIs** are the **Google Maps API** and **OpenStreetMap**.

Example:

- You type in: "Empire State Building, New York."
- Geo Encoding translates it to: approximately **40.7484∘ N, 73.9857∘ W**. The map then places a pin at that exact location.

Spatial Data Attributes

- Every item on a digital map (like a point, line, polygon, or grid cell) already has its location (its (x,y) or latitude/longitude coordinates).
- Attributes are the non-location data that describe the feature.
- **Example:**
  - **Altitude (or z value):** For a point on the Earth's surface, the attribute could be its height above sea level (its elevation).
  - **Names:** A spatial object (like a river or a city) can have a name attached to it.

Choropleth

- A choropleth is a map where defined areas (like counties, states, or countries) are colored or shaded based on a specific statistic.
- **Example:** A darker shade of blue might represent a higher rate of unemployment in a county, while a lighter shade of blue represents a lower rate.
- Choropleth maps are excellent tools for visualizing and communicating how a statistic is distributed or varies across a region.
- They allow for quick, visual comparison between neighboring geographic units.

Coding

Mapping in R

```
install.packages("maps")
install.packages("ggmap")

library(ggplot2)
library(maps)
states <- map_data("state")
head(states)

##        long      lat group order  region subregion
## 1 -87.46201 30.38968     1     1 alabama      <NA>
## 2 -87.48493 30.37249     1     2 alabama      <NA>
## 3 -87.52503 30.37249     1     3 alabama      <NA>
## 4 -87.53076 30.33239     1     4 alabama      <NA>
## 5 -87.57087 30.32665     1     5 alabama      <NA>
## 6 -87.58806 30.32665     1     6 alabama      <NA>
```

install.packages("maps"):

- This **installs** the R package called "**maps**" from the internet.
- "**maps**" contains map data like countries, states, and regions.

install.packages("ggmap")

- Installs the "**ggmap**" package.
- "ggmap" lets you create maps using **ggplot2**, and add layers like points, lines, and labels on maps.

states <- map_data("state")

- **map_data("state")** gets the **map of U.S. states**.
- **states <-** stores this map data in a variable called **states.**

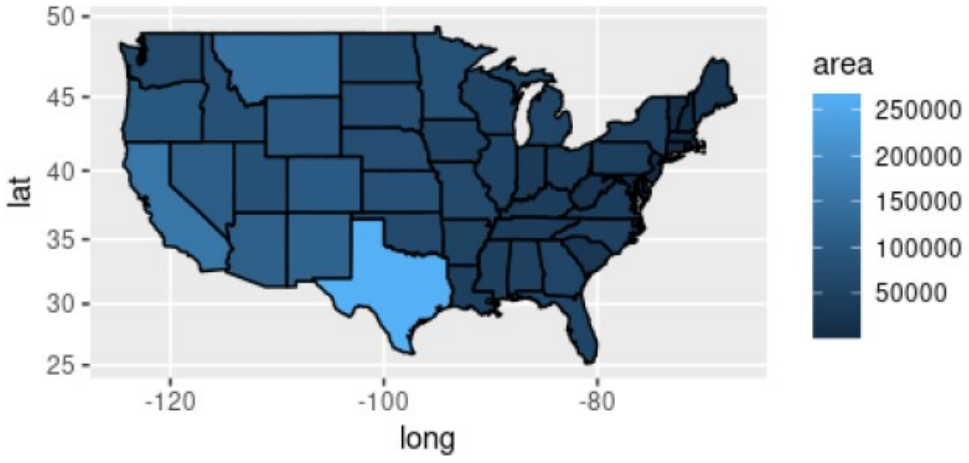
•This helps you **peek at what's inside** before plotting.

```r
ggplot(states)  + geom_point(aes(x=long,y=lat)) + coord_map()
```

- **ggplot(states)**: This starts the entire plot-making process and specifies the data to be used.
- **geom_point(aes(x=long, y=lat))**: This tells ggplot2 how to draw the data.
- **geom_point():** Specifies the geometric object to use, in this case, points (dots).
- **coord_map()**: This applies a map projection to the plot to account for the Earth's spherical shape."
- **Without coord_map(),** the map would look distorted and squashed because the raw latitude and longitude coordinates are simply plotted on a flat, rectangular plane, which doesn't correctly represent the **true distances and shapes of the states**.

Creating a Filled Map (Based on Area):

```
ggplot(usDataWithGeom) +
    geom_polygon(color="black",
                 aes(x=long,y=lat, group=group,  fill=area)) +
    coord_map()
```



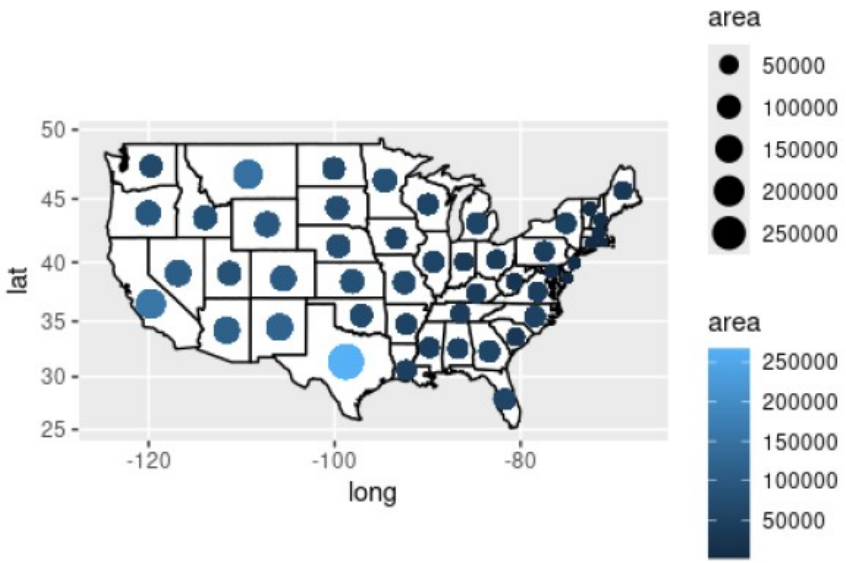
- **ggplot(usDataWithGeom): usDataWithGeom** has map coordinates (**long and lat**) for U.S. states and extra data like **area.**

geom_polygon(color="black", aes(x=long, y=lat, group=group, fill=area)):

- **geom_polygon()** → Draws shapes (polygons) by connecting points.
- **group=group** → Ensures the points are connected in the right order for each state.
- **fill=area** → The inside color of each state depends on the value of area. States with bigger/smaller area will have different colors.
- **color="black"** → Draws a black border around each state.

Add Points to the Map

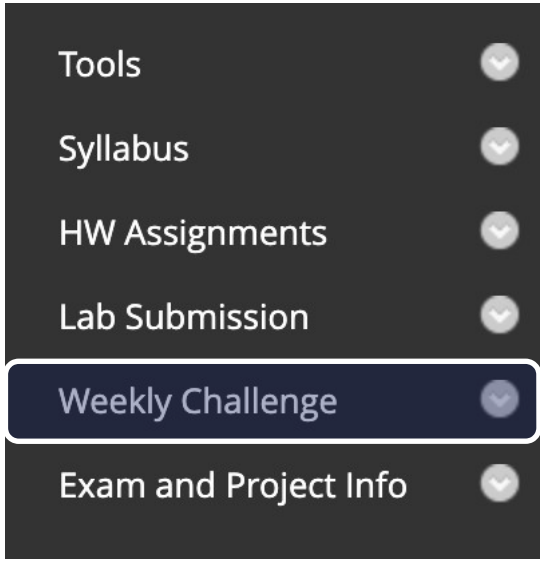
```
ggplot(usDataWithGeom) +
  geom_polygon(color="black", fill="white",
               aes(x=long,y=lat, group=group)) +
  geom_point(
      aes(x=centerX,y= centerY, color=area,size=area)) +
  coord_map()
```



geom_point(aes(x=centerX, y=centerY, color=area, size=area)):

- Adds a point in the middle of each state (using centerX, centerY).
- The color of the point depends on the value of area.
- The size of the point also depends on area.

Weekly Quiz


Tools
Syllabus
HW Assignments
Lab Submission
Weekly Challenge
Exam and Project Info

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The End